2.3 Counting Curve Measurements

NOTE: The system is very sensitive to changes in high voltage and you should allow the system to stabilize (~ 30 s) after changing the voltage before measuring the count rate. Ensure that your counting curve begins below the lowest voltage that generates visible pulses from the output of the amplifier, which can be observed on the scope. Make sure the high voltage is varied over a range such that all the features of the counting curve can be observed. Collect enough data points to distinguish the features of the counting curve. Set the counting time be long enough that uncertainty in counts is low. **DO NOT EXCEED 1700 V**.

2.3.1 Generation of the counting curve using an alpha source

- First, we will determine the count rate vs. voltage characteristic of the counter using a source that only emits alphas, such as ^{241}Am.
- Record the gross counts exceeding the LLD channel for several applied voltages. The applied voltages should range from ~300V to 800V (make sure to take more counts in the knee region).
  - You can do this by selecting the ROI to be the chosen LLD channel to the maximum channel. The gross counts will then be the counts above the LLD
  - Alternatively, you can select all channels, right click, and choose "Sum".
- Record the peak centroid at each high voltage setting
  - You can do this by estimating the peak centroid or reselecting the ROI to be the peak region
- Display the amplifier outputs on the oscilloscope during these measurements and be alert for the point at which the alpha pulses begin to saturate.
- Record a sample differential pulse height spectra.

2.3.2 Generation of the counting curve using a beta source

- In this section the applied voltage range should be from ~1000V up to 1700V. Repeat section 2.2.3 so that at the maximum voltage, the Beta spectrum can still be fully observed.
  - You will need to choose a new LLD and Gain following the steps in 2.2.3
- Repeat the measurements described in Section 2.3.1 using a source that emits betas but no alphas (such as ^{60}Co or ^{90}Sr). For this section, there is no need to record the peak centroid.

3 Absolute Counting Efficiency

3.1 Measure the absolute counting efficiency of the source-chamber geometry using a calibrated alpha or beta reference source. Compare your measurement to the maximum absolute efficiency you would expect for this counter.

- Select an optimal operating voltage and LLD based on previous measurements so that all source counts can be recorded
- Take a 1 minute measurement and use the count rate to determine the current activity as well as the original source activity.
  - Use the date marked on the source to calculate the original activity
  - Compare your calculated rate with the original source activity.
  - How should the rate you measure be related to the marked activity?

4 Gamma Response

Observe the counter response to gamma rays only by using the ^{60}Co or ^{137}Cs source turned upside down. In this manner, the betas are blocked, but the higher energy gamma rays will penetrate into the counter gas.

- Repeat the measurements described in Sections 2.2.3 and 2.3.2
  - Again, for this section do not record the centroid channel (no peak will be visible).